INCH-POUND
MS27786D
w/AMENDMENT 1
23 January 2014
SUPERSEDING
MS27786D
18 April 2011

DETAIL SPECIFICATION SHEET

# SWITCH, TOGGLE, TWO POLE, ENVIRONMENTALLY SEALED, INTEGRATED WIRE TERMINALS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switch described herein shall consist of this specification and the latest issue of MIL-DTL-3950.

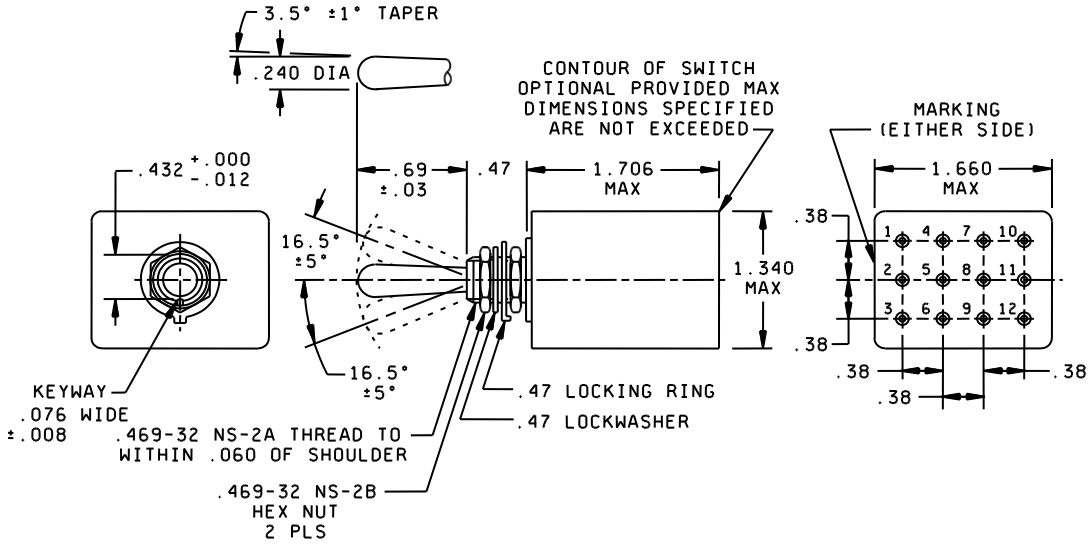



FIGURE1. Dimensions and configuration



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:51Z
Check the source to verify that this is the current version before use.

| Inches | mm | Inches | mm |
|---|---|---|---|
| .005 | 0.13 | .38 | 9.7 |
| .008 | 0.20 | .432 | 10.97 |
| .012 | 0.30 | .469 | 11.91 |
| .020 | 0.51 | .47 | 11.9 |
| .03 | 0.8 | .69 | 17.5 |
| .060 | 1.52 | 1.340 | 34.04 |
| .076 | 1.93 | 1.660 | 42.16 |
| .240 | 6.10 | 1.706 | 43.33 |

NOTES:

1. Dimensions are inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .020 (0.51 mm) for two place decimals and ± .005 (0.13 mm) for three place decimals.
4. For hardware detail specifications see appendix of MIL-DTL-3950.
5. Contact installing/ removal tool in accordance with MIL-I-81969/14-02.
6. Grommet sealing plug, in accordance with MS27488-A20.
7. Terminal shall adequately accept a wire contact within dimensional limits of SAE-AS39029/1-101.
8. In the event of a conflict between the text of this standard and the reference cited herein, the text of this standard shall take precedence.

FIGURE1. Dimensions and configuration Continued



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:51Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

The Sealing grommet shall seal on a smooth wire insulation of .040 (1.02 mm) to .083 (2.11mm) diameters

The terminal end of switch shall be color coded red to indicate contact size

Direction of internal mechanism movement is opposite to the direction of the toggle movement

Sealing plugs may be used in nonfunctional grommet holes.

Maximum weight is .21pound.

Electrical rating: See table I.

TABLE I: Detail Requirements

| MS Part Number | Circuit with toggle lever in | | | Current capacity (amperes) 28 volts DC | | | Current capacity (amperes) 115 volts, 60 and 400 hertz AC | | |
|---|---|---|---|---|---|---|---|---|---|
| | Keying Side | Center | Opposite keying side | Lamp-load circuit | Resistive circuit | Inductive circuit | Lamp-Load circuit | Resistive circuit | Inductive circuit |
| MS27786-1 | 1-2 4-5<br>7-8 ON 10-11 | 2-3 4-5<br>7-8 ON 11-12 | 2-3 5-6<br>8-9 ON 11-12 | 5 | 7.5 | 7.5 | 4 | 7.5 | 7.5 |
| MS27786-2 | 1-2 MOM. 4-5<br>7-8 ON 10-11 | 2-3 4-5<br>7-8 ON 11-12 | 2-3 5-6<br>8-9 ON 11-12 | 4 | 7.5 | 7.5 | 2 | 7.5 | 7.5 |
| MS27786-3 | 1-2 MOM. 4-5<br>7-8 ON 10-11 | 2-3 4-5<br>7-8 ON 11-12 | 2-3 MOM. 5-6<br>8-9 ON 11-12 | | | | | | |
| MS27786-21 | 1-2 7-8<br>4-5 ON 10-11 | OFF | 2-3 8-9<br>5-6 ON 11-12 | 5 | 7.5 | 7.5 | 4 | 7.5 | 7.5 |
| MS27786-22 | OFF | NONE | 2-3 8-9<br>5-6 ON 11-12 | | | | | | |
| MS27786-23 | 1-2 7-8<br>4-5 ON 10-11 | NONE | 2-3 8-9<br>5-6 ON 11-12 | | | | | | |
| MS27786-24 | NONE | OFF | 2-3 8-9<br>5-6 ON 11-12 | | | | | | |
| MS27786-25 | NONE | MOM. OFF | 2-3 8-9<br>5-6 ON 11-12 | 4 | 7.5 | 7.5 | 2 | 7.5 | 7.5 |
| MS27786-26 | 1-2 MOM. 7-8<br>4-5 ON 10-11 | NONE | 2-3 8-9<br>5-6 ON 11-12 | | | | | | |
| MS27786-27 | 1-2 MOM. 7-8<br>4-5 ON 10-11 | OFF | 2-3 MOM. 8-9<br>5-6 ON 11-12 | | | | | | |
| MS27786-28 | 1-2 MOM 7-8<br>4-5 ON 10-11 | OFF | NONE | | | | | | |
| MS27786-29 | MOM. OFF | NONE | 2-3 8-9<br>5-6 ON 11-12 | | | | | | |
| MS27786-30 | 1-2 MOM. 7-8<br>4-5 ON 10-11 | NONE | OFF | | | | | | |
| MS27786-31 | 1-2 MOM. 7-8<br>4-5 ON 10-11 | OFF | 2-3 8-9<br>5-6 ON 11-12 | | | | | | |
| MS27786-32 | NONE | 1-2 7-8<br>4-5 ON 10-11 | 2-3 MOM 8-9<br>5-6 ON 11-12 | | | | | | |
| MS27786-33 | NONE | 1-2 7-8<br>4-5 ON 10-11 | 2-3 8-9<br>5-6 ON 11-12 | 5 | 7.5 | 7.5 | 4 | 7.5 | 7.5 |

Test Requirements

Test shall be performed in accordance with MIL-DTL-3950 except

a) During all tests. Switches shall be fully wired with appropriate wire and terminal contacts.
b) Contact voltage drop- the contact drop with two terminals and the switch contact in series shall not exceed 8 millivolts measured from one wire contact through the contacts to the other wire contacts.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:51Z
Check the source to verify that this is the current version before use.

Referenced documents
MIL-DTL-3950

The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity
DLA – CC

(Project 5930-2011-139)

Review activities
Army – AR, AV, MI
Navy – EC, MC
Air Force – 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T16:51Z
Check the source to verify that this is the current version before use.